WORKED SOLUTION

STEP I 1999, Question 1

Pure • Counting Coprime Integers

> **STEP I 1999, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/99-s1-q1

The key idea

Divisibility by 2 and 5 depends only on the last digit, and divisibility by 3 and 7 depends only on the remainder modulo 21. So the integers we want repeat in a fixed pattern, block after block, and counting them is a matter of counting one block and multiplying. For the averages, the trick is symmetry: if n is in the set then so is its "partner" $N - n$, so the numbers pair off around the midpoint.

Solution

Part 1: not divisible by 2 or 5, below a million

"Not divisible by 2 or 5" means divisible by neither. An integer is divisible by 2 or by 5 exactly when its last digit is $0, 2, 4, 5, 6$ or 8. So we want the integers whose last digit is 1, 3, 7 or 9.

The integers $0, 1, \ldots, 999\,999$ split into $100\,000$ blocks of ten consecutive integers $10k, 10k+1, \ldots, 10k+9$, and each block contains exactly four of the integers we want. Hence the count is

$$4 \times 100\,000 = 400\,000.$$

(Inclusion–exclusion gives the same: $10^6 - \frac{1}{2}10^6 - \frac{1}{5}10^6 + \frac{1}{10}10^6 = 400\,000$.)

For the average, pair each such n with $10^6 - n$. If n ends in $1, 3, 7, 9$ then $10^6 - n$ ends in $9, 7, 3, 1$ respectively, and $0 < 10^6 - n < 10^6$ because $n \neq 0$. So $n \mapsto 10^6 - n$ maps the set to itself, and it is its own inverse. It has no fixed point, since $n = 500\,000$ is even. So the set splits into $200\,000$ pairs, each with sum 10^6. The total is $200\,000 \times 10^6$, and the average is

$$\frac{200\,000 \times 10^6}{400\,000} = 500\,000.$$

Part 2: not divisible by 3 or 7, below 4179

The number 4179 is chosen with care:

$$4179 = 3 \times 1393 = 3 \times 7 \times 199 = 21 \times 199.$$

Whether n is divisible by 3 or by 7 depends only on the remainder of n on division by 21. Among the remainders $0, 1, \ldots, 20$, the multiples of 3 are $0, 3, \ldots, 18$ (seven of them) and the multiples of 7 are $0, 7, 14$ (three), with 0 counted in both. So $21 - 7 - 3 + 1 = 12$ remainders are divisible by neither:

$$1,\ 2,\ 4,\ 5,\ 8,\ 10,\ 11,\ 13,\ 16,\ 17,\ 19,\ 20.$$

The integers $0, 1, \ldots, 4178$ make up exactly 199 complete blocks of 21, so the count is

$$12 \times 199 = 2388.$$

As a check, by inclusion–exclusion: $4179 - 1393 - 597 + 199 = 2388$.

> **Ask yourself** Why does the symmetry trick still work here, when the numbers have nothing to do with decimal digits?

The key is that 21 divides 4179. Then $4179 - n \equiv -n \pmod 3$ and $\pmod 7$, so $3 \mid 4179 - n$ if and only if $3 \mid n$, and the same for 7. Also $n = 0$ is excluded (it is divisible by everything), so every n in the set has $1 \le n \le 4178$, and then $1 \le 4179 - n \le 4178$ too. So $n \mapsto 4179 - n$ maps the set to itself. It has no fixed point because 4179 is odd. The 2388 integers therefore form 1194 pairs, each summing to 4179, and the average is

$$\frac{1194 \times 4179}{2388} = \frac{4179}{2} = 2089\tfrac{1}{2}.$$

Checks

A computer that runs through all the integers confirms the counts 400 000 and 2388 and the averages 500 000 and 2089.5 exactly. You can also do a small case by hand: below 21, the twelve remainders listed above add up to $126 = 12 \times \frac{21}{2}$, so their average is $10.5 = \frac{21}{2}$, exactly as the pairing argument predicts.

Common slips

- Reading "not divisible by 2 or 5" as "not divisible by both", i.e. not divisible by 10. That would give 900 000 integers.
- Forgetting to add back the multiples of 21 (or of 10) in inclusion–exclusion, since they have been subtracted twice.
- Assuming the average must be the midpoint of the range without checking that the set is symmetric. It works here only because 10 divides 10^6 and 21 divides 4179.

Beyond STEP

The count of integers in $1, \ldots, m$ coprime to m is Euler's totient function $\varphi(m)$. Here $\varphi(10) = 4$ and $\varphi(21) = 12$, and the counts are $\varphi(10^6) = 400\,000$ and $199\,\varphi(21)$. The pairing $n \leftrightarrow m - n$ shows that the integers coprime to m always add up to $\frac{1}{2} m\,\varphi(m)$ for $m > 2$, a fact that is a first step into the arithmetic of the integers modulo m. Davenport's *The Higher Arithmetic* is an excellent place to read more.

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